

Enhancing flexural performance of rubberized concrete beams through incorporation of rice husk ash as cement replacement

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ABSTRACT

The quest for sustainable concrete production emphasizes human health and environmental safety while significantly reducing CO₂ emissions associated with traditional cement manufacturing, which is vital in combating climate change. Moreover, the novel application of by-products with pozzolanic properties not only diminishes environmental waste but also enhances the performance of concrete constructions. This research studies the sustainable application of crumb rubber (CR) and rice husk ash (RHA) in reinforced concrete (RC) beams. The research proposes a sustainable approach to mitigate the decreased flexural capacity in rubberized RC beams by incorporating rice husk ash (RHA) as a substitute for cement. Nine different concrete mixtures were designed with varying substitution ratios of RHA and CR as partial replacements for cement and fine aggregate (FA), respectively. The substitution ratios considered were 0 %, 10 %, or 20 % for RHA and 0 %, 5 %, or 10 % for CR by FA. Nine simply supported RC beams were cast and tested. The findings demonstrated that substituting cement with RHA at an ideal replacement ratio of 10 % enhanced the concrete strength, whereas the use of CR had a detrimental impact. The flexural capacity, stiffness, and toughness of the examined beams improved with the inclusion of RHA up to 10 %; they then declined with the inclusion of 20 % RHA but continued to be higher than those of the control specimen. The exploration of concrete mixture substitution ratios highlights a compelling balance between performance and sustainability. According to the experimental results, the best substitution ratio for the concrete mixture is a combination of 10 % RHA and 10 % CR, which only slightly reduces the ultimate load by 3.1 %. Finally, the experimental outcomes were compared with the predicted results of ACI 318–19 approaches.

1. Introduction

Our world is grappling with two significant issues: the first is the depletion of natural resources, and the second is pollution [1–4]. Concrete is the most-used material in construction. Alternative materials that can replace conventional materials are essential for its making, as it requires a large amount of aggregate and cement. Moreover, awareness of using waste materials and by-products is increasing. waste tires Rubber is a non-biodegradable material. Traditional methods of disposing of waste tire rubber include burning, landfilling, or dumping. This method not only leads to environmental and health issues but also contributes to the annual increase in the number of discarded vehicle

tires. Therefore, as an environmentally friendly alternative, researchers are using crumb rubber (CR) to partially replace fine aggregate (FA) in concrete [5–7]. Studies around the world are carried out on the usage of supplementary cementitious materials (SCMs) [8–11], since cement production results in the release of a large amount of carbon and greenhouse gases. Currently, the use of agricultural wastes is gaining popularity. During milling, the protective covering of a rice grain separates into rice husk ash (RHA). The usage of RHA can improve the mechanical properties of concrete, reduce costs, and lower the carbon footprint [12–15].

Considering that approximately a ton of greenhouse gases are emitted for every ton of cement produced [16], cement production

Abbreviations: B, Beam breadth mm; f_c' , cylinder compressive strengths MPa, in the current paper $f_c' = 0.8 f_{cu}$; A_s , Tension area of steel; ϵ_s , maximum steel strain; C_c , Compression force in the concrete compression zone; f_{tc} , Tensile strength; D, Effective depth mm; f_y , yield stress of main steel; A_s' , Compression area of steel; T_s , Tension force in the tension steel bars; T_c , Compression force in the compression steel bars; D', Concrete cover.

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significantly contributes to greenhouse gas emissions and poses serious environmental challenges. So, introducing RHA as an SCMs markedly reduces the carbon footprint associated with cement production by decreasing the quantity of cement needed. CO₂ emissions and energy consumption are the two main criteria taken into account by the environmental assessment. Both of these factors influence the environmental impact of cement manufacture. The CO₂ emissions from burning RHA are almost 20 % of the CO₂ emissions from cement production, according to Hu et al. [17]. Utilizing RHA is sustainable since, according to reports, the energy required to burn it is hardly ever considered in comparison to the energy required to create cement. Many studies have explored the influence of RHA on the properties of fresh and hardened concrete. Due to its porous nature, with both macro- and meso-pores on and within the particles, RHA has a high water absorption capacity, resulting in a significantly large specific surface area. This can reduce workability and increase the potential for segregation in the concrete mixture. To mitigate these effects and maintain the desired workability, superplasticizers are commonly added. This helps to ensure that the concrete remains cohesive and workable despite RHA's tendency to absorb large amounts of water [18,19]. RHA generally improves the mechanical properties of concrete, with consistent results demonstrating an increase in strength as RHA concentration increases until a specific limit is reached, after which it begins to decrease. According to [20], the optimal RHA content is 10 %, while [21] suggests an optimum of 15 %, and [22] found that RHA improves the flexural strength of high-strength concrete up to a 20 % ratio. Several studies [23–26] have explored the durability of concrete incorporating RHA as a SCM. The fine particles of RHA contribute to a dense microstructure, acting as inhibitors and enhancing the concrete's resistance to acids, sulfates, chlorides, alkali-silica reactions, and carbonation. Furthermore, RHA greatly reduces drying shrinkage and thermal conductivity. Similarly, research [27,28] focused on the shear behavior of RHA on RC beams showed a gain in strength as RHA content increases until a certain limit, at which point it begins to deteriorate. Both studies identified 10 % as the optimal RHA ratio for shear strength. As a result, the excess silica remains unreacted, leading to a decrease in concrete strength [29–31]. Zeyad et al. [32] and Shaban et al. [33] came to the conclusion that basil plant ash and sugar beet ash are suitable replacements for ordinary Portland cement in the production of ultra-high-performance concrete with excellent properties. Amin et al. [34] studied the use of peanut shell ash and sunflower shell ash as partial substitutes for cement in the production of sustainable high-strength concrete. The findings revealed that the ideal substitution rate for cement is 10 % peanut shell or 30 % sunflower shell ash by weight of cement to get good mechanical properties. Zeyad et al. [35] found that adding 20 % of olive waste ash to high-strength geopolymer concrete mixtures instead of fly ash resulted in a substantial improvement in compressive strength.

The use of CR in construction not only reduces the environmental problems associated with tire waste but also contributes to the reduction of carbon footprints by utilizing a renewable resource [36,37]. Its use is seen as a crucial step in reducing the environmental burden associated with tire disposal while also addressing broader sustainability goals [38]. Several researchers [39–45] have extensively investigated the workability and mechanical properties of rubberized concrete. They have observed that the workability of rubberized concrete decreases as the CR content increases, primarily due to heightened friction between the rubber particles and the cement paste. Furthermore, incorporating CR into concrete results in reductions in compressive strength, flexural strength, splitting tensile strength, and modulus of elasticity. Many studies [46–49] have examined the durability of rubberized concrete and found that adding CR can enhance its freeze-thaw resistance, as rubber particles act similarly to traditional air-entraining agents. Additionally, CR improves acid and sulfate resistance, reduces chloride ion penetration, and lowers thermal conductivity. At the level of concrete elements, several investigations were conducted. Hossain et al. [50] focused on the shear behavior of rubberized beams, which also indicates

a consistent reduction in strength with increasing CR content. Ahmed et al. [51] stated that adding treated rubber and waste tire steel fibers greatly increased the shear strengths of the concrete. Furthermore, Elbially et al. [52] and Aly et al. [53] concluded that the flexural performance of rubberized beams also experiences a consistent decline in strength with increased CR content. The researchers [54,55] have explored the impact of CR content on the punching shear strength of rubberized concrete slabs, with results consistently showing a decline in strength as CR content increases. The authors [56,57] reported that as the CR amount and heating temperatures increased, the columns' capacity decreased. The quantity of CR influenced the reduction in column capacity more significantly than the heating temperatures. Elsayed et al. [58] investigated the effects of using waste glass powder as a cement substitute to improve the axial capacity of rubberized concrete columns. The incorporation of GP up to 20 % as a cement substitute significantly affected the column's capacity.

Prior research determined that substituting cement with RHA enhances concrete strength, whereas the incorporation of CR adversely affects both concrete strength and the structural behavior of RC beams. This paper's novelty lies in evaluating the combined effect of RHA and CR as substitutes for cement and FA, respectively, on the flexural performance of RC beams.

2. Research significance

By utilizing recycled materials like CR and RHA, the study aims to not only lower CO₂ emissions but also promote sustainability by minimizing waste and conserving resources. Consequently, this approach contributes to reducing the overall ecological footprint of conventional concrete production. The primary objectives of this study are bifurcated into two stages: the first stage involved examining the mechanical characteristics of concrete mixtures incorporating a combination of CR and RHA as FA and cement, respectively. The next stage involved examining the effect of using RHA in improving the flexural performance of rubberized concrete beams.

3. Experimental work

3.1. Material

The used raw materials in the experimental work were ordinary Portland cement (OPC), RHA, coarse aggregate, fine aggregate, CR, superplasticizer, and tap water. OPC (CEM I 42.5 N) have a specific surface area of 360 m²/Kg, and a specific gravity of 3.15 and met the requirements of ASTM C150 [59]. RHA with particle sizes ranging from 5 to 12 µm, a particle size range-specific surface area of 287 m²/Kg, and a specific gravity of 2.07, was used as a partial substitute for OPC. According to ASTM C618 [60], RHA qualifies as a pozzolanic material. The production process involves burning the raw rice husks in a controlled furnace at approximately 700°C for five hours before allowing them to cool to room temperature. The coarse aggregate used in this study was crushed dolomite, which was washed with potable water to ensure it was free of dust. In accordance with ASTM C127, the coarse aggregate had a grading range between 4.75 mm and 22.4 mm, with a specific gravity of 2.70, a bulk density of 1.65 t/m³, a fineness modulus of 2.98, and a water absorption rate of 1.30 %. Natural sand was used as FA, with particle sizes ranging from 0.18 mm to 2.8 mm. The properties of the fine aggregate were evaluated based on ASTM C128 [61], revealing a specific gravity of 2.65, a bulk density of 1.69 t/m³, a fineness modulus of 2.5, and a water absorption rate of 1.73 %. CR was used as a partial replacement for the FA. To eliminate the zinc stearate coating formed on the rubber surface during tire manufacturing, the rubber underwent a pre-treatment process. The rubber was soaked in a 10 % sodium hydroxide (NaOH) solution to enhance its bonding with the cement matrix and improve its mechanical properties. The pretreatment involved cleaning the rubber with tap water to remove any dirt, immersing it in a

10 % NaOH solution for 1 h, and then stirring it in water until the pH dropped from 14 to 7. Afterward, the rubber was left to air-dry before mixing [39,62]. The pre-treatment procedure is illustrated in Fig. 1. The properties of the CR after treatment were determined according to ASTM C128 [61]. The CR exhibited a grading range from 0.18 mm to 1.40 mm, a specific gravity of 0.96, a bulk density of 0.61 t/m^3 , a fineness modulus of 2.02, and a water absorption rate of 0.32 %. A sieve analysis of the aggregates was performed following ASTM C136 [63]. Potable water was used to mix the concrete. A superplasticizer (SP) (Sika ViscoCrete 3425) was introduced to ensure that the concrete mixture achieved the requisite workability. For the reinforcement, three types of steel bars were used: 8 mm diameter for stirrups with a yield strength of 218 MPa, 10 mm diameter bars with a yield strength of 522 MPa for the upper longitudinal bars, and 12 mm diameter bars with a yield strength of 578 MPa for the lower longitudinal bars. The properties of the steel were tested according to ASTM A370 [64].

3.2. Concrete mixture design

Nine concrete mixtures were created according to ACI 211.1 [65], featuring different combinations that included partial replacement of OPC with RHA at ratios of 0 %, 10 %, or 20 %. Additionally, FA was partially replaced with CR at ratios of 0 %, 5 %, or 10 %. Table 2 outlines the specific proportions of these concrete mixtures. Previous studies indicated that adding CR or RHA to concrete mixtures reduced their workability [15,18,39,44,45]. Several tests with different amounts of superplasticizer were done, as shown in Table 2, to keep the slump values for concrete mixtures at $150 \text{ mm} \pm 10 \text{ %}$.

3.3. Test program

Nine RC beams, one for each concrete mixture, were cast and tested. Each beam had dimensions of 120 mm (width), 250 mm (height), and 2000 mm (length). The steel reinforcements were identical across all beams, with the lower reinforcements consisting of $2\Phi 12$ mm bars and the upper reinforcements consisting of $2\Phi 10$ mm bars. The stirrups utilized consisted of two branches, each measuring approximately 8 mm at 100 mm, and were covered with a 25 mm concrete cover. Details, dimensions, and reinforcement of the beams are shown in Fig. 2.

3.4. Test setup

The tested beams were simply supported with a total length of 2000 mm and an effective span of 1800 mm. To enhance crack visibility during and after testing, the beam surfaces were painted white beforehand. As shown in Fig. 3, a 600 kN hydraulic jack, operating at a rate of 0.25 mm/min, applied a vertical load (P) to the beam. This load was distributed across two points along the span of the tested beam through two points. The central deflections of the tested beams were measured with a linear variable differential transducer (LVDT) with a 100-mm stroke. A data acquisition system was linked to each instrument for

data collection.

4. Result and discussion

4.1. Mechanical characteristics of concrete mixtures

4.1.1. Compressive strength

According to EN 12390-3 [66], Fig. 4 and Table 3 show the compressive strengths of different concrete mixtures after 28 days. The results demonstrated that adding 10 % RHA resulted in a remarkable improvement in compressive strength of 19.6 %, respectively, compared to conventional concrete. It may be due to RHA comprising a considerable quantity of silica (SiO_2), which, when mixed with lime (calcium hydroxide, Ca(OH)_2) from cement, undergoes a pozzolanic reaction. This reaction produces additional calcium silicate hydrate (C-S-H), which is the main compound responsible for the strength and durability of concrete. The formation of C-S-H improves the bond between particles and enhances the material's mechanical properties, including flexural strength. When 20 % RHA replaced OPC, the compressive strength decreased, yet it remained 6.9 % higher than the reference mix. The extra silica that RHA provides turns into an inert substance and doesn't contribute to strength building when its content increases from 10 % to 20 %. This is because the hydration process doesn't produce enough C-H to react with all of the silica that RHA makes available [67,68]. The additional silica provided by RHA can be identified as an inert material that does not aid in strength development. This occurs due to the insufficient quantity of C-H produced during the hydration process, which is inadequate to react with all the silica introduced by the RHA. The authors [27,29,69] observed similar outcomes. CR notably has a stronger effect in reducing compressive strength. However, 5 % and 10 % CR reduced compressive strength by 4.2 % and 23.9 %, respectively. The decrease in the compressive strength of concrete that includes a mixture of CR can be attributed to CR's low stiffness, which allows cracks to propagate more easily under loading. Similar outcomes were observed by the authors [54,55]. The addition of RHA to rubberized concrete can be mitigating the negative effect caused by the introduction of CR. It can be explained by CR creating weak zones in concrete due to its lower stiffness compared to the cement matrix. RHA reduces these weak zones by enhancing the cohesion and uniformity of the matrix. The fine particles of RHA act as micro-fillers, reducing the voids within the concrete matrix. A denser concrete matrix ensures better confinement of the steel reinforcement and higher overall capacity. The combined use of 5 % CR and 10 % RHA positively influences concrete compressive strength. Adding 10 % RHA and 5 % CR improved compressive strength by 14 %. Whereas 10 % CR and 10 % RHA resulted in a 6.4 % decrease compared to conventional concrete. Additionally, incorporating 20 % RHA and 5 % CR reduced compressive strength by 3.2 %, while using 10 % CR led to a 17 % decrease compared to conventional concrete.

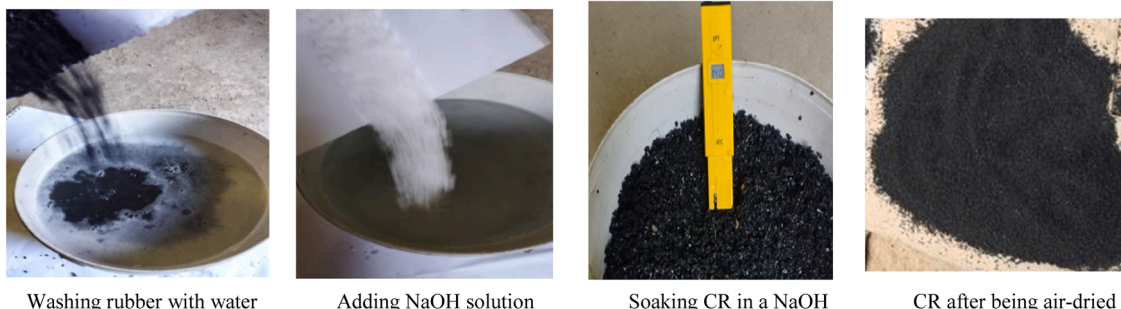


Fig. 1. Rubber particle pretreatment procedure.

Table 1

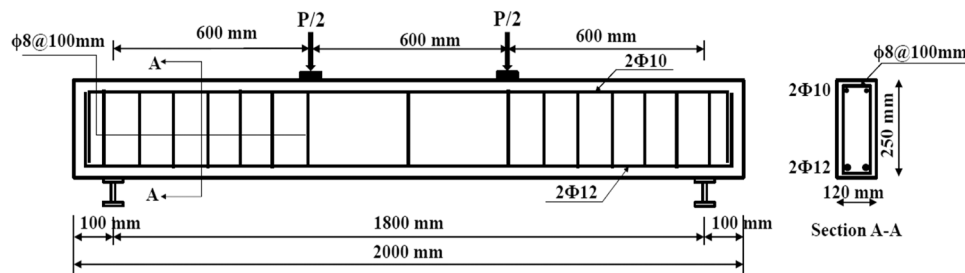
The chemical composition of RHA and OPC.

Chemical Oxides (%)	SiO ₂	SO ₃	MgO	Na ₂ O	Fe ₂ O ₃	CaO	Al ₂ O ₃	K ₂ O	LOI	CL	TiO ₂	P ₂ O ₅
RHA	77.62	2.46	0.73	0.06	3.67	11.77	1.11	1.96	5.12	0.47	0.19	1.18
OPC	20.81	1.27	1.51	0.40	1.38	64.36	4.75	0.56	4.42	-	-	-

Table 2Mixture proportions of concrete (kg/m³).

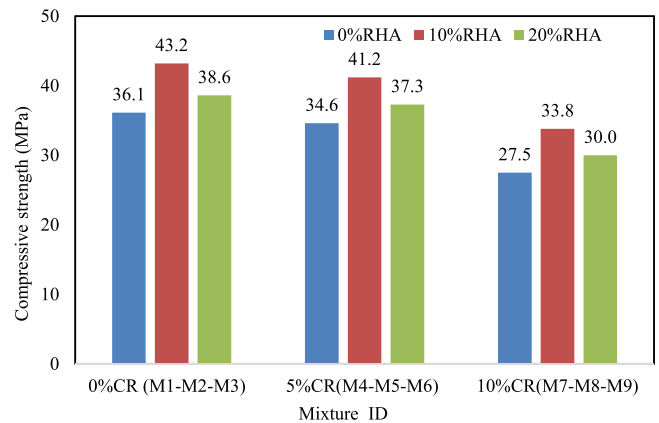
ID	Mixture Code	OPC	RHA	% RHA	CA	FA		% CR	water	SP
						FA	CR			
M1*	CR0-RHA0	400	0	0	1205	602	0	0	190	1.20
M2	CR0-RHA10	360	40	10	1205	602	0	0	190	1.50
M3	CR0-RHA20	320	80	20	1205	602	0	0	190	2.00
M4	CR5-RHA0	400	0	0	1205	572	30	5	190	2.25
M5	CR5-RHA10	360	40	10	1205	572	30	5	190	3.50
M6	CR5-RHA20	320	80	20	1205	572	30	5	190	4.50
M7	CR10-RHA0	400	0	0	1205	542	60	10	190	3.25
M8	CR10-RHA10	360	40	10	1205	542	60	10	190	4.50
M9	CR10-RHA20	320	80	20	1205	542	60	10	190	5.40

* Control mixture

**Fig. 2.** Dimensions and details of reinforcement for the test beams.**Fig. 3.** Test setup.

4.1.2. Splitting tensile strength

Fig. 5 and Table 3 display the splitting tensile strength of different concrete mixtures after 28 days of curing, as per ASTM C496 [70]. The inclusion of RHA resulted in an enhanced tensile strength, with 10 % and 20 % RHA increasing tensile strength by 13.3 % and 3.3 %, respectively, compared to conventional concrete. This increase is likely due to the pozzolanic activity of RHA particles, which act as filler material within capillary pores. In contrast, CR exhibited a more pronounced negative impact on tensile strength than RHA, with 5 % and 10 % CR reducing tensile strength by 6.6 % and 26.6 %, respectively. The weak and smooth texture of rubber particles may have contributed to the reduction in splitting tensile strength by facilitating the rapid propagation of cracks under loading. The combination of 10 % RHA and 5 % CR improved tensile strength by 6.6 % relative to conventional concrete. The inclusion of RHA may have enhanced the interfacial bond between CR and the cement paste. Better load transfer in the concrete

**Fig. 4.** Compressive strength of concrete mixtures.

results from the enhanced ITZ's reduction of stress concentrations and prevention of premature cracking. Overall, mixtures containing 10 % CR tended to show a decline in tensile strength, with the reduction becoming more substantial as the RHA contents increased. Specifically, adding 10 % or 20 % RHA with 10 % CR resulted in decreases of 6.6 % and 20.0 %, respectively, relative to conventional concrete. The authors observed similar trends in other studies [23,27,54,55].

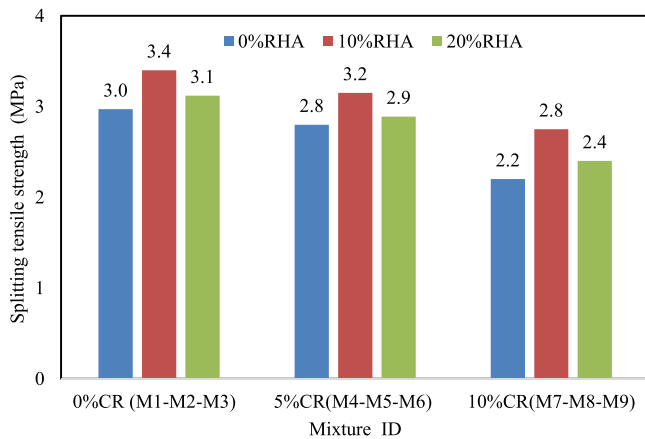
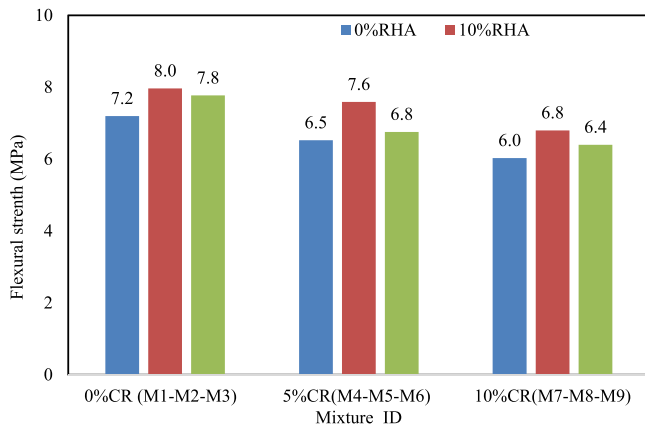
4.1.3. Flexural strength

Fig. 6 and Table 3 display the 28-day flexural strength of different concrete mixtures. The results show that adding 10 % RHA resulted in an improvement in flexural strength of 11.1 % compared to the control

Table 3

Compressive, tensile, and flexural strengths of concrete mixtures.

Mix #		*M1	M2	M3	M4	M5	M6	M7	M8	M9
Compressive Strength (MPa)	Individual	37.0	45.0	38.6	33.0	41.5	40.1	30.3	32.6	28.6
		37.1	40.4	40.6	35.1	44.0	38.2	26.1	33.0	30.6
		34.2	44.1	36.6	35.7	38.1	33.4	26.2	35.9	30.9
	Average	36.1	43.2	38.6	34.6	41.2	37.3	27.5	33.8	30.0
	SD	1.6	2.4	2.0	1.4	2.9	3.4	2.4	1.8	1.2
	COV	4.5 %	5.6 %	5.3 %	4.0 %	7.1 %	9.2 %	8.6 %	5.3 %	4.1 %
Tensile Strength (MPa)	Individual	2.9	3.7	3.0	2.7	3.1	2.6	2.4	2.7	2.3
		3.1	3.2	3.0	2.7	3.4	3.3	2.2	2.7	2.5
		3.1	3.2	3.3	3.0	3.1	2.9	2.0	3.0	2.4
	Average	3.0	3.4	3.1	2.8	3.2	2.9	2.2	2.8	2.4
	SD	0.1	0.3	0.2	0.1	0.2	0.3	0.2	0.1	0.1
	COV	4.1 %	8.6 %	5.3 %	5.3 %	6.4 %	11.7 %	8.7 %	5.3 %	4.7 %
Flexural Strength (MPa)	Individual	7.5	7.6	8.3	6.9	7.3	6.5	5.7	7.2	6.1
		6.7	8.1	7.9	6.5	8.2	6.6	6.4	6.8	6.5
		7.5	8.2	7.2	6.1	7.4	7.2	5.9	6.4	6.6
	Average	7.2	8.0	7.8	6.5	7.6	6.8	6.0	6.8	6.4
	SD	0.5	0.3	0.6	0.4	0.5	0.4	0.4	0.4	0.3
	COV	6.3 %	4.1 %	7.1 %	5.5 %	6.4 %	5.3 %	5.9 %	5.5 %	4.1 %

**Fig. 5.** Splitting tensile strength of concrete mixtures.**Fig. 6.** Flexural strength of the tested beams.

mixture. It may be due to The use of RHA enhances the pore structure of concrete by reducing both the number and size of the pores. Denser and less porous concrete may more effectively disperse applied stresses, boosting its capacity to resist bending and enhancing flexural strength. The flexural strength decreased with the addition of 20 % RHA, but it remained higher than that of the control mixture by about 8.3 %. This can be attributed to surpassing the quantity of accessible lime in the mixture, resulting in incomplete pozzolanic reactions. This may result in

a reduction in the overall strength of the concrete due to the production of calcium silicate hydrate (C-S-H). Conversely, CR has an opposite effect on flexural strength, as increments of 5 % and 10 % CR alone cause decreases of 9.7 % and 6.9 %, respectively. The CR particles may act as stress concentrators or microcracks within the concrete matrix. This can lead to premature failure under flexural loads, as the material may fracture more easily at the rubber-particle interfaces, resulting in reduced strength. The combination of 10 % RHA and 5 % CR resulted in a 5.6 % increase in flexural strength, while the addition of 10 % CR led to a 5.6 % decrease compared to conventional concrete. The addition of 20 % RHA with 5 % and 10 % CR resulted in a 5.6 % decrease in flexural strength. The authors observed similar outcomes [27,54,55].

4.2. Results of the tested beams

4.2.1. Failure modes and cracking pattern

Fig. 7 illustrates the failure modes and crack patterns of the tested beams. Each beam experienced a ductile collapse, characterized by the standard flexural failure mode. Initial cracks appeared as the load increased, developing near the bottom chord of the beam (tension side), nearly vertical to the beam's center. As the load continued to rise, existing cracks expanded, and new ones formed and spread. In the final loading stages, these cracks extended, with wider cracks reaching the top chord of the beam (compression side). In some beams, partial cover separation occurred, and the rate of new crack formation significantly slowed. The addition of RHA was found to delay crack initiation and control crack width and spread, likely due to enhanced tensile strength and a denser concrete microstructure. As anticipated from previous studies, the inclusion of CR had an adverse effect on the cracking of the beams. Rubberized concrete beams exhibited earlier and more frequent crack formation. CR also contributed to increased crack widths, which grew with higher CR levels, likely due to the smooth surface of rubber particles. This, in turn, led to weak adhesion between the rubber and cement paste, promoting the development of additional cracks under load. Table 4 shows that the first cracking load of the tested beams varied between 44.7 % and 64.1 % of the failure load.

4.2.2. Load-deflection curves

Fig. 8 presents the load-deflection curves for the examined beams. These curves exhibit three distinct phases. The initial phase, known as the elastic stage, exhibits linear behavior from the onset of loading until the first crack load. In the second stage, upon reaching the initial crack load, the beams' load-deflection behavior exhibits nonlinearity as a result of crack formation. In the final phase, all beams attain their maximum load, resulting in a ductile failure. The findings demonstrate that an increase in RHA content within concrete mixtures results in a

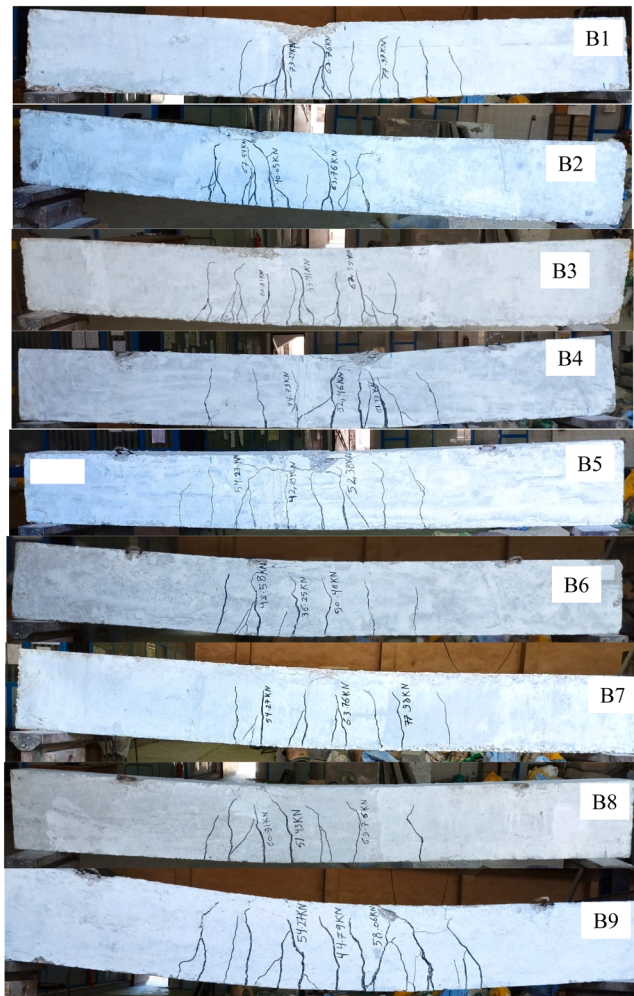


Fig. 7. Failure modes and cracking pattern in tested beams at failure.

decrease in ultimate deflection. The incorporation of 10 % and 20 % RHA led to a reduction in ultimate deflection of 8.6 % for beam B2 and 13.6 % for beam B3, relative to the control beam B1. This impact may arise from the pozzolanic reaction of RHA particles, which enhances the concrete's pore structure and augments the matrix's density. Consequently, the concrete is more adept at resisting deformation [71–74]. In contrast, incorporating CR led to a considerable rise in the ultimate deflection of rubberized RC beams compared to the control beam. This is due to the low elasticity modulus of CR, which leads to the formation of more cracks when the structure is loaded, compared to the control beam.

The obtained results were comparable with outcomes concluded by the authors [54,55,62]. Combining CR and 10 % RHA reduces the ultimate deflection compared to beams without RHA. The ultimate deflection was roughly 8.7 % and 4.3 % less for rubberized concrete beams with 10 % RHA and 5 % or 10 % CR, respectively, compared to the control beam (B1). Similarly, the combination of CR and 20 % RHA significantly decreases ultimate deflection in comparison to mixtures without RHA. The ultimate deflection of rubberized concrete beams containing 20 % RHA and 5 % and 10 % CR was approximately 13.7 % and 8.7 % lower, respectively, compared to control beams (B1).

4.2.3. Flexural capacity

Table 4 demonstrates that incorporating RHA improves the flexural capacity of the test specimens, with the most significant improvement observed at 10 % RHA. Although the flexural capacity decreased with 20 % RHA, it continued to be higher than that of the control beam. Beams B2 and B3 showed increases of 11.4 % and 2.70 % in flexural capacity with the addition of 10 % and 20 % RHA, respectively, compared to the control beam. This improvement is probably due to the pozzolanic reaction of RHA particles, which makes the concrete matrix stronger and helps the concrete and reinforcement bars stick together better. The finer particle size of RHA improves aggregate interlock; hence, it enhances flexural capability. In contrast, adding CR had a negative effect on the flexural capacity of the tested beams, with this deterioration increasing as the CR level rose. The incorporation of 5 % and 10 % CR resulted in a flexure capacity that was approximately 6.2 % and 18.5 % lower than the control beam, respectively. This could be due to the lower stiffness, higher porosity, lower quality, and inconsistent composition of CR, as well as its weaker bonding properties, which reduced the strength of the concrete and made it more difficult for the aggregates to fit together. These factors lead to reduced structural integrity, especially under bending or flexural stresses. The author [52, 53] came to similar conclusions. The incorporation of 10 % RHA in rubberized concrete beams reduces the adverse impact associated with the inclusion of CR. The addition of RHA contributes to filling voids between CR and cement particles, resulting in a denser and more compact microstructure. RHA enhances the interfacial transition zone (ITZ) by refining pore structure and diminishing micro-crack dimensions. Compared to the control beam, the incorporation of 10 % RHA with 5 % and 10 % CR (B5 and B6) improved the flexural capacity by approximately 6.5 % and 0.60 percent, respectively. Combining CR with 20 % RHA resulted in a flexural capacity that was approximately 3.2 % and 13.8 % lower for 5 % and 10 % CR, respectively, compared to the control beam. The use of CR in RC beams primarily results in a decrease in flexural strength due to higher porosity, a poorer and variable composition of CR, higher shrinkage, and a weaker bond between the CR aggregates and the cement paste. Enhancing CR processing may be necessary to enhance the performance of RC beams incorporating CR.

Table 4
Experimental results of the tested beams.

Beams #	Mixture Code	First cracking load (kN)	Deflection at first cracking load (mm)	Ultimate load (kN)	Ultimate deflection (mm)	Initial Stiffness (kN/mm)	Toughness (kN. mm)
*B1	CR0-RHA0	63.8	6.50	102.5	11.50	10.8	1181.0
B2	CR0-RHA10	66.7	5.50	114.2	10.00	16.8	1552.4
B3	CR0-RHA20	66.5	5.50	105.3	9.50	14.2	1451.9
B4	CR5-RHA0	50.6	5.00	96.2	12.00	10.7	1349.9
B5	CR5-RHA10	67.5	6.00	109.2	10.50	16.5	1352.7
B6	CR5-RHA20	64.5	5.50	102.2	10.00	14.0	1444.9
B7	CR10-RHA0	36.6	4.50	83.5	12.5	9.1	1167.1
B8	CR10-RHA10	48.2	4.50	99.3	11.00	12.4	1494.5
B9	CR10-RHA20	42.8	4.50	88.3	10.50	11.5	1284.9

* Control Beam

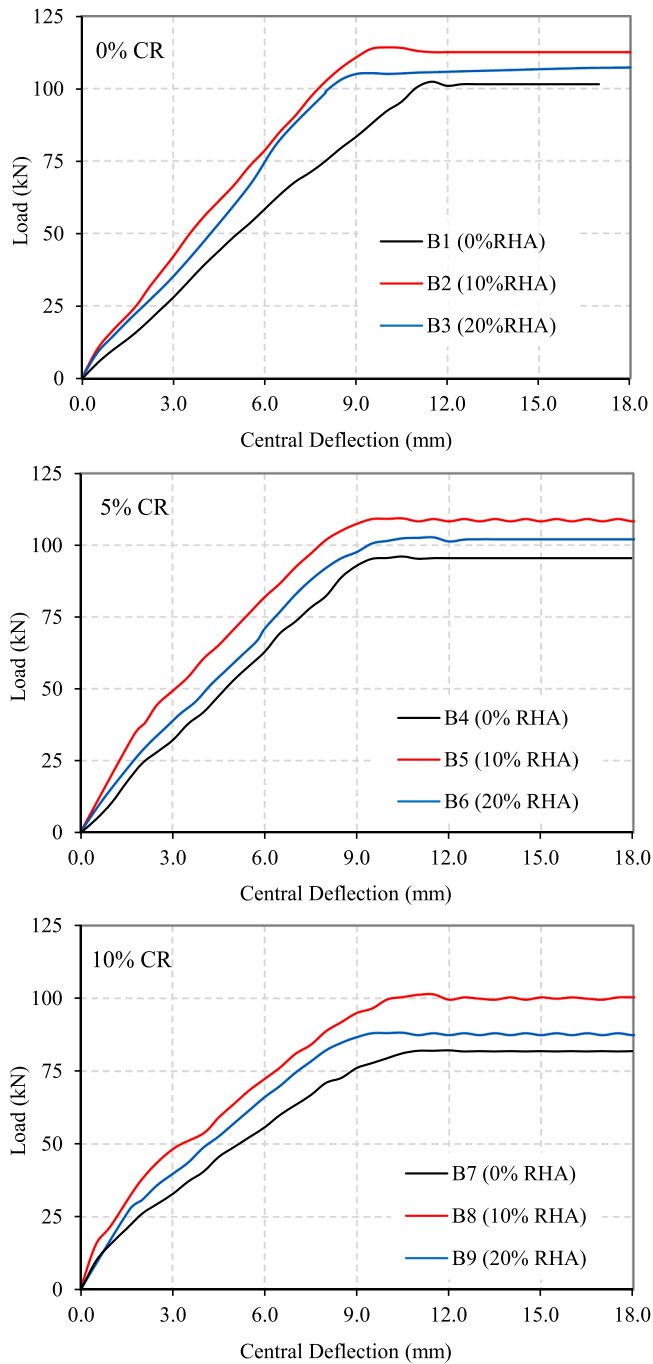


Fig. 8. Load-deflection curves of tested beams.

4.2.4. Stiffness

The load deflection curves were used to calculate the initial stiffness, which quantifies resistance to deformation under an applied force. The incorporation of RHA increased the stiffness of the examined beams, as indicated in Table 4. The pozzolanic reaction of RHA particles, which filled in capillary pores, probably helped make the concrete stronger, improved its microstructure, and reduced micro-cracking, which made it stiffer. Incorporating 10 % and 20 % RHA increased the stiffness of the beams (B2 and B3) by 55.5 % and 31.5 %, respectively, compared to the control beam. The addition of 5 % CR caused a slight reduction in the stiffness of the beam (B4) by about 0.92 %. The addition of 10 % CR to the beam (B7) resulted in a 15.7 % reduction in stiffness compared to the control beam (B1). The low stiffness and modulus of elasticity of CR cause a direct reduction in the stiffness of the tested beams by promoting

the formation of additional cracks under load. It was observed that the incorporation of RHA can mitigate the negative effect due to the addition of CR. The addition of RHA enhances the bond with reinforcement and improves the strength loss due to CR. The combination of 10 % RHA with 5 % and 10 % CR improved stiffness by 52.7 % and 29.6 %, respectively, compared to the control beam.

4.2.5. Toughness

Toughness is defined as the area below the load-deflection curve. Table 4 displays the toughness values computed based on the test results. The results indicate that adding up to 10 % RHA significantly improves the toughness of the tested beams. Although toughness decreased slightly at 20 % RHA, it still surpassed that of the control beam. Specifically, 10 % RHA increased toughness by 31.4 %, while 20 % RHA led to a 22.9 % increase compared to conventional concrete. This likely happens because of the pozzolanic reaction of RHA particles, which fill in the spaces between the capillaries and make the concrete stronger by improving its microstructure and lowering micro-cracking. All of these things make the concrete tougher. However, a 5 % CR increase in toughness is 14.3 %, whereas a 10 % CR reduces it by 1.2 % when compared to conventional concrete. The low toughness and modulus elasticity of CR cause a direct decrease in the toughness of the tested beams. This effect is due to the influence on the formation of new cracks during the loading process. The combination of 10 % RHA and 5 % CR increased toughness by 14.5 %, but 10 % CR increased toughness by 26.5 %. Incorporating 20 % RHA with 5 % and 10 % CR led to a 22.3 % and 8.8 % increase in toughness strength, respectively, compared to ordinary concrete.

5. Assessment of the flexure strength design code formulas

In this investigation, the flexural capacity of the test specimens was predicted using the American Building Code ACI-318 (2019) [75]. The objective of this point is to assess the congruence between the design code formulas and the experimental study outcomes. Fig. 9 illustrates the strain distribution and net tensile strain in accordance with ACI-318 (2019) [75]. The predicted flexural capacity can be determined using Eqs. 1 through 9.

Take $\epsilon_{cr} = 0.1\epsilon_0$ assume $\epsilon_0 = 0.003 \therefore \epsilon_{cr} = 0.0003$

$$\frac{x}{\epsilon_{cr}} = \frac{d-c}{\epsilon_s} \therefore x = \frac{\epsilon_{cr}}{\epsilon_s} (d-c) \quad (1)$$

$$\frac{y}{10\epsilon_{cr}} = \frac{d-c}{\epsilon_s} \therefore y = \frac{10\epsilon_{cr}}{\epsilon_s} (d-c) \quad (2)$$

$$C_c + T_c = T_s + f_{tc} \therefore y = 10 * x \quad (3)$$

$$C_c = 0.85 * f_c' * a * b \quad (4)$$

$$T_c = A_s' * f_y \quad (5)$$

$$T_s = A_s * f_y \quad (6)$$

$$M_{U_{internal}} = [T_s * \left(d - \frac{a}{2}\right) + T_c * \left(\frac{a}{2} - d'\right) + f_{tc} * b * (d - a)] * 10^{-6} \quad (7)$$

$$M_{U_{internal}} = M_{U_{external}} \quad (8)$$

$$2 * P = \frac{2 * M_{U_{external}}}{a_{shear \ span}} \quad (9)$$

5.1. Comparison of experimental results and predictions for ACI-318-19 [75] design code

Fig. 10 compares the ratio of the experimental ultimate load to the predicted ultimate load per ACI 318-19 [75]. The ratios vary across the

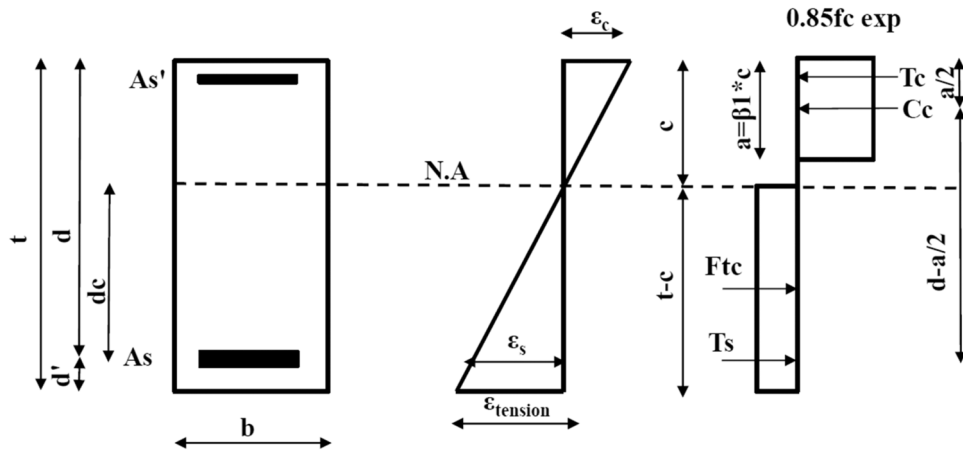


Fig. 9. Equivalent rectangular stress block according to ACI-318 [75].

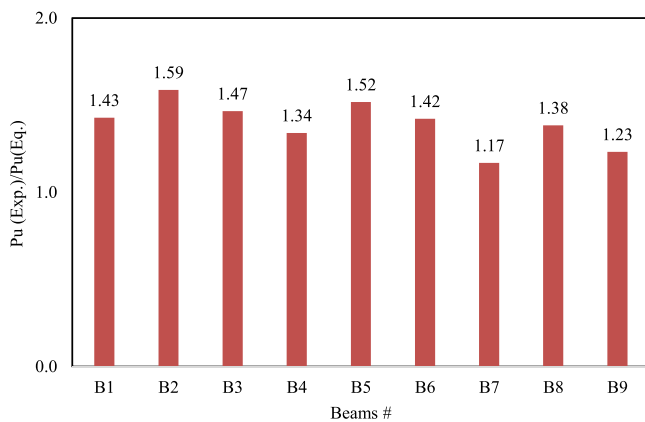


Fig. 10. The ratio between experimental and ACI equation results.

beams, ranging from 1.17 (lowest for beam B7) to 1.59 (highest for beam B2). The average, standard deviation, and coefficient of variation of the ratio $P(\text{Exp.}) / P(\text{Eq.})$ are 1.39, 0.13, and 6.8 %, respectively. For the tested beams, the experimental ultimate load is consistently greater than the predicted ultimate load, as indicated by the majority of values being above 1.0. Although there is a considerable difference among the beams, the experimental ultimate loads typically surpass the expected values. This implies that the flexural capacity estimates provided by the prediction model are conservative. Experimental results may deviate from forecasts because of differences in material properties, beam construction, or simplifications in the model.

6. Conclusions

This research presents the experimental results for nine RC beams incorporating RHA and CR that were subjected to flexural tests. The results encompass the mechanical properties of the concrete mixtures as well as the flexural behavior of the tested beams. Based on these experimental outcomes, the following conclusions are drawn within the range of the studied parameters:

1. The incorporation of RHA enhanced the compressive, tensile, and flexural strengths of concrete mixtures. Conversely, the addition of CR has a negative impact, while the addition of 10 % RHA can mitigate this effect. The concrete strengths progressively decreased as the amounts of RHA and CR increased.
2. The addition of RHA improves the flexural capacity of the tested beams, with the biggest ratio recorded at 10 % RHA. Although the

flexural capacity decreased with 20 % RHA, it endured greater than that of the control beam. The flexural capacity of the tested beams incorporating 10 % and 20 % RHA was 11.4 % and 2.70 %, respectively, higher than that of the control beam.

3. Adding CR weakened the beam's strength, and the amount of CR used directly affected the lower flexural strength; this impact can be lessened by adding 10 % RHA.
4. The addition of CR and RHA improved the stiffness and toughness of rubberized concrete beams. The stiffness and toughness of the tested beam, which contained 10 % CR and 10 % RHA, were higher than those of the control beam by about 29.6 % and 22.3 %, respectively.
5. The exploration of concrete mixture substitution ratios highlights a compelling balance between performance and sustainability. By incorporating 10 % RHA and 10 % CR, the resulting concrete achieves an acceptable ultimate load, stiffness, and toughness (albeit with a modest 3.1 % drop) and champions environmental sustainability in construction practices. This innovative approach to mixture design emphasizes the potential for alternative materials to enhance structural integrity while minimizing ecological impact, paving the way for more sustainable building solutions that can meet contemporary engineering demands.
6. The ACI 318 design code equations tend to be conservative when estimating the flexural performance of beams.

Recommendations for future research

The authors recommend additional investigation into rubberized concrete beams' flexural performance with differing CR and RHA substitution ratios. The authors recommend conducting statistical and numerical analyses to develop a reliable modeling methodology. This will assist in determining the key elements that affect the performance of RC beams containing CR and RHA.

CRediT authorship contribution statement

Alturki Mansour: Writing – review & editing, Formal analysis. **Elsayed Mahmoud:** Writing – review & editing, Supervision, Methodology. **Abou Elmaaty Mohamed:** Writing – review & editing, Supervision. **Hussein Noha:** Writing – original draft, Software, Investigation, Formal analysis.

Declaration of Competing Interest

We pledge that we are not involved in the interests of the financial, commercial, legal, or professional relationship with other organizations, or with the people we worked with them, that could influence my research

Data availability

Data will be made available on request.

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